

Nishant Chandna

+91 9625148396 | chandna.nishant@gmail.com | linkedin.com/in/nishant | github.com/nishant

EDUCATION

Delhi Technological University

B.Tech in Mechanical Engineering with Specialization in Automotive Engineering

August 2023 – May 2027

9.42 CGPA

HONORS AND AWARDS

- **2nd Position (Self-Funded Category)** in the **DARPA (Defense Advanced Research Projects Agency) Triage Challenge Phase-2**, winning a prize of **\$150,000**.
- **2nd Position (Self-Funded Category)** in the **DARPA (Defense Advanced Research Projects Agency) Triage Challenge Phase-1**, winning a prize of **\$60,000**.
- **2nd Position** in the simulation stage and **3rd place** in the physical stage of **ICUAS (International Conference on Unmanned Aircraft Systems) 2024**.
- **Research Excellence Award** by Delhi Technological University

EXPERIENCE

Vice Captain & Autonomy Lead | UAS-DTU

August 2023 – Present

- Led the Autonomy Team for the DARPA Triage Challenge, designing and implementing autonomy frameworks that enable robots to navigate, explore, and make decisions in unknown and dynamic environments.
- Developed and integrated perception, planning, and control modules for coordinated multi-robot behavior across both ground and aerial platforms.

Voluntary Researcher | SAIR Lab, University of Buffalo

May 2026 – Present

- Working on Vision-Language Navigation (VLN) for embodied AI applications, focusing on frontier-based exploration and long-horizon autonomous navigation in unknown environments. Developing multimodal perception and decision-making pipelines integrating vision, language, and spatial reasoning for robotics systems.

Founding Research Intern | FPV Labs, Bangalore

May 2026 – Present

- Developing egocentric pose estimation and multimodal perception solutions for robotics applications, focusing on temporal motion modeling and long-sequence consistency. Working on real-world embodied AI and visual SLAM pipelines for autonomous systems

Research Intern | RRC, IIIT Hyderabad (Prof. Sourav Garg)

Feb 2026 – May 2026

- Working on navigation based on topological maps generated from multi-view image segmentation and inter-frame data association, constructing region-level graphs for efficient graph-based planning and localization.

Research Intern | RRC, IIIT Hyderabad (Prof. Madhava Krishna)

May 2025 – July 2025

- Implemented degeneracy awareness and localizability analysis to improve robustness in LiDAR odometry and mapping.
- Integrated Scan Context as a backend for place recognition in state-of-the-art ICP algorithms, adding loop closures and factor graph optimization for global consistency in SLAM pipelines, reducing global drift by 40%.

PUBLICATIONS

Plann3r: Predicting Planning Costs Grounded in 3D

2026

CORL 2026 (Under Review)

- 3D grounded method to predict pixel-level planning cost for any given set of images with an arbitrary subgoal pixel.
- Integrated learned costmap prediction with control-conditioned planning, outperforming connectivity-based search planners across diverse navigation tasks.

DAMM-LOAM: Degeneracy-Aware Multi-Metric LiDAR Odometry and Mapping

2025

IROS Active Perception Workshop 2025

- Proposed a novel framework to improve pose estimation in challenging, low-feature environments such as corridors.
- Demonstrated improved performance over state-of-the-art LOAM frameworks in both degenerate and outdoor scenarios.

Autonomous UAV Navigation and Mapping for Accurate Fruit Detection and Counting

2025

International Conference on Unmanned Aircraft Systems (ICUAS)

- Presented a UAV-based fruit counting framework combining YOLOv8 detection, 3D mapping via LiDAR-camera fusion, and TSP-A*-based path planning.
- Achieved 98% accuracy in simulation and 90% in real-world testing; placed 2nd in simulation and 3rd overall at ICUAS'24 UAV Competition.

PROJECTS

Drone based forest mapping | *C++, Python, Bash, PX4, ROS, Git*

- Deployed a PX4-based UAV in a simulated forest environment in Gazebo; implemented forest mapping using GenZ-ICP and benchmarked it against KISS-ICP; fused ICP outputs with GPS and IMU data for robust state estimation using dynamic confidence-based weighting.
- Designed and simulated custom environments to systematically evaluate and validate the mapping and localization pipeline.

Collision Avoidance using Ego Planner and Vins Mono | *C++, VIO, PX4, ROS*

- Implemented GPS-denied autonomous navigation and collision avoidance in indoor environments using VINS-Mono for visual-inertial state estimation and EGO-Planner for real-time local trajectory generation.
- Integrated perception, planning, and control modules on a lightweight quadrotor running PX4 and ROS, enabling agile flight in cluttered 3D spaces with high-speed obstacle avoidance.

Reward Function Improvement in PyFlyt (RL for UAVs) | *RL, Stable Baselines 3, Gymnasium*

- Contributed to a reinforcement learning environment repository by enhancing the reward function, introducing a negative reward based on the rate of change of orientation over previous time steps to improve drone stability in scenarios like hovering and pole balancing. Significantly improving efficiency and convergence speed.

TECHNICAL SKILLS

Hardware: Jetson Nano, Jetson Orin, Arduino, Pixhawk, LiDARs, Intel RealSense Tracking Camera

Software: ROS, PX4, Gazebo, ArduPilot SITL, Docker, Git

Programming Languages: Python, C++, C, C#

Robotics: SLAM, Sensor Fusion, Path Planning, LIO, VIO, UAV Autonomy, Reinforcement Learning